21-242 Recitation 12

Rank–nullity — Q1. (T A)= (A)

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Prove that $\dim(\mathrm{im}(T_A)) = \mathrm{rank}(A)$.

Course-level solution

By definition, $\mathrm{im}(T_A)$ is the column space of A, and $\mathrm{rank}(A)$ is defined to be the dimension of the column space (equivalently, the number of pivot columns of A, which form a basis of that column space by RREF Q8). Hence the equality is tautological once those identifications are made.

If rank was defined as the number of nonzero rows of $\mathrm{RREF}(A)$: that number equals the number of pivot columns, which is the length of a basis of $\mathrm{im}(T_A)$.